Quantification: 535_z27c1-2
==========================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	23
interior lacuna count	19
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 19)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3183.79	2123	2922.77	px^2
major_axis_length_px	lacuna shape	89.497	70.3776	42.3512	px
minor_axis_length_px	lacuna shape	46.0965	43.1713	21.7406	px
aspect_ratio	lacuna shape	2.0321	2.1668	0.7351	unitless
eccentricity	lacuna shape	0.8092	0.8871	0.1376	unitless
solidity	lacuna shape	0.7997	0.7988	0.0959	unitless
roots_count	headline	7.3684	6	4.2974	count
ring_length_r30_px	headline	234.105	208	114.252	px
ring_length_r60_px	reported	515.684	457	257.619	px
owned_length_px	ownership-dependent	615.338	300.836	614.424	px
edge_count	ownership-dependent	27.2632	13	25.6901	count
mean_edge_length_px	ownership-dependent	22.6204	22.7446	7.0041	px
perimeter_px	normalised	292.206	250.392	153.2	px
ring_area_r30_px2	normalised	9256.42	7991	4202.49	px^2
ring_area_r60_px2	normalised	19809.9	18793	8568.03	px^2
in_frame_fraction_r30	normalised	0.993	1	0.0304	unitless
in_frame_fraction_r60	normalised	0.9866	1	0.0547	unitless
ring_density_r30	normalised	0.0256383	0.0248781	0.00716352	px^-1
ring_density_r60	normalised	0.0262262	0.0255106	0.00723202	px^-1
roots_per_100px_perimeter	normalised	2.6185	2.325	1.1559	per 100 px
ring_attached_length_r30_px	network v2	194.474	168	117.043	px
ring_attached_length_r60_px	network v2	337.105	244	240.723	px
ring_length_w_r30_px	network v2	265.07	230.095	129.052	px
ring_length_w_r60_px	network v2	583.803	522.002	295.83	px
sholl_crossings_r10	network v2	6.2105	5	3.9802	count
sholl_crossings_r20	network v2	7	5	3.7712	count
sholl_crossings_r30	network v2	7.3684	7	4.0443	count
loop_count	loops, report only	1.2105	0	2.2005	count
loop_edge_share	loops, report only	0.0694	0	0.1183	share
ring_width_mean_r30_px	width	7.4563	7.7652	1.7003	px

Per field
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measure	value	units
analysed_area_px2	973979	px^2
total_skeleton_length_px	24236	px
canalicular_length_density_per_px	0.0248835	px^-1
junction_count	436	count
junction_density_per_px2	0.00044765	px^-2
lacuna_count	23	count

lacunae_per_px2	2.361e-05	px^-2
skeleton_component_count	402	count
mean_component_length_px	60.2886	px
median_component_length_px	23	px
field_length_density_w_per_px	0.0280678	px^-1
field_density_without_flagged_per_px	0.0233411	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	1242	px
unattached_component_count	75	count
canalicular_length_density_attached_per_px	0.0248835	px^-1
field_length_density_with_unattached_per_px	0.0261587	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	32	count
loop_edge_share	0.0947	share
edge_count	1256	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.6103	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	18913	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2942	101.888	45.2188	2.2532	0.8961	0.7542	-1.1126	105.425	685.744	False
2	810	40.5726	31.6629	1.2814	0.6253	0.7226	0.1823	107.814	615.768	False
3	957	54.4409	24.7864	2.1964	0.8903	0.7988	0.0308	119.021	790.769	False
4	1484	55.1992	46.0237	1.1994	0.5521	0.6199	-0.2191	163.815	808.798	False
5	955	64.3571	19.6453	3.276	0.9523	0.8459	1.0779	145.542	454.49	False
6	3599	102.364	47.1738	2.1699	0.8875	0.8955	-1.0474	233.848	107.369	False
7	1807	63.8235	38.7415	1.6474	0.7947	0.9372	1.5189	218.143	826.646	False
8	712	35.4872	28.8379	1.2306	0.5828	0.9368	-0.3495	301.131	1012.33	True
9	11051	179.07	107.127	1.6716	0.8013	0.7079	0.7029	439.405	253.828	False
10	5312	120.404	59.815	2.0129	0.8679	0.8459	-1.268	460.68	557.419	False
11	1257	48.9366	34.4065	1.4223	0.7111	0.929	1.2464	562.555	134.776	False
12	796	41.9622	30.4426	1.3784	0.6882	0.761	0.6839	561.568	949.678	False
13	2123	56.2494	49.4219	1.1381	0.4775	0.9222	0.3101	574.703	493.107	False
14	2990	94.2448	43.1713	2.183	0.8889	0.8771	0.9473	600.532	860.511	False
15	1556	70.3776	32.2729	2.1807	0.8887	0.7988	1.1709	626.462	984.626	False
16	1310	65.9258	27.032	2.4388	0.9121	0.8501	-1.1711	624.014	302.034	False
17	9402	173.447	80.0495	2.1668	0.8871	0.7022	-0.8455	691.761	675.898	False
18	2231	116.941	28.0885	4.1633	0.9707	0.7914	0.9981	758.602	307.963	False
19	6401	142.513	65.0238	2.1917	0.8898	0.8153	-1.2115	832.616	623.668	False
20	3509	107.726	65.7294	1.6389	0.7923	0.6187	-0.7908	851.443	291.978	False
21	1656	75.1349	31.8764	2.3571	0.9055	0.792	-0.7612	861.306	996.914	True
22	754	37.3455	26.9997	1.3832	0.6909	0.9286	-0.1889	961.662	1011.9	True

23 10983 260.5 56.6496 4.5984 0.9761 0.8482 -1.5452 998.074 728.234 True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	6	280	535	792.097	46	17.2195	9.5449
2	9	218	540	264.309	21	12.5861	2.8284
3	4	133	236	136.953	7	19.5648	8.3237
4	7	208	337	255.924	11	23.2658	8.0191
5	2	107	344	70.0833	4	17.5208	
6	6	284	834	1790.48	84	21.3152	7.1459
7	4	140	363	101.154	4	25.2886	7.8006
8	2	92	276	116.841	7	16.6915	7.7797
9	15	445	1043	1159.97	51	22.7446	7.7299
10	18	336	754	935.754	48	19.4949	8.3125
11	7	161	457	525.002	17	30.8825	8.5498
12	3	84	144	37.2132	5	7.4426	3.9876
13	5	180	387	295.794	9	32.866	8.4103
14	10	245	465	215.551	12	17.9626	6.5239
15	6	148	290	300.836	12	25.0696	9.9005
16	3	142	269	147.154	4	36.7886	7.8455
17	13	492	1008	2077.65	78	26.6366	7.5315
18	4	190	407	331.35	13	25.4885	7.2625
19	8	270	641	755.813	39	19.3798	7.3389
20	10	385	744	1498.33	53	28.2704	7.1584
21	4	121	213	126.154	6	21.0257	7.0065
22	2	77	153	125.811	5	25.1622	8.2557
23	12	408	906	1149	74	15.5271	7.4854

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	369.203	9004	16772	1	1	0.0310973	0.0318984	1.6251
2	188.102	5174	12677	1	1	0.0421337	0.0425968	4.7846
3	159.095	4461	7968	1	1	0.0298139	0.0296185	2.5142
4	254.072	5329	9525	1	1	0.0390317	0.0353806	2.7551
5	163.723	7120	19836	1	1	0.0150281	0.0173422	1.2216
6	281.948	10538	26647	1	1	0.0269501	0.0312981	2.128
7	184.853	6087	14824	1	1	0.0229998	0.0244873	2.1639
8	112.426	3564	9950	0.5858	0.5585	0.0258137	0.0277387	1.7789
9	645.151	19717	40885	1	1	0.0225694	0.0255106	2.325
10	368.818	12630	27408	1	1	0.0266033	0.0275102	4.8805
11	147.054	6972	18793	1	1	0.0230924	0.0243176	4.7602
12	153.338	5451	11094	1	0.9843	0.01541	0.01298	1.9565
13	184.267	7991	18400	1	1	0.0225253	0.0210326	2.7135
14	250.392	9403	19497	1	1	0.0260555	0.0238498	3.9937
15	219.35	5949	9604	0.8676	0.7611	0.0248781	0.0301958	2.7354
16	195.48	7849	18629	1	1	0.0180915	0.0144399	1.5347
17	634.625	17491	33867	1	1	0.0281288	0.0297635	2.0485
18	284.534	9498	19681	1	1	0.0200042	0.0206798	1.4058
19	465.843	13368	26300	1	1	0.0201975	0.0243726	1.7173
20	402.073	11840	23982	1	1	0.0325169	0.0310233	2.4871
21	212.551	6244	13037	0.731	0.6597	0.0193786	0.0163381	1.8819
22	116.805	3670	7367	0.6042	0.5352	0.0209809	0.0207683	1.7123
23	816.873	12795	26681	0.621	0.5937	0.0318875	0.0339568	1.469

Network v2 per lacuna

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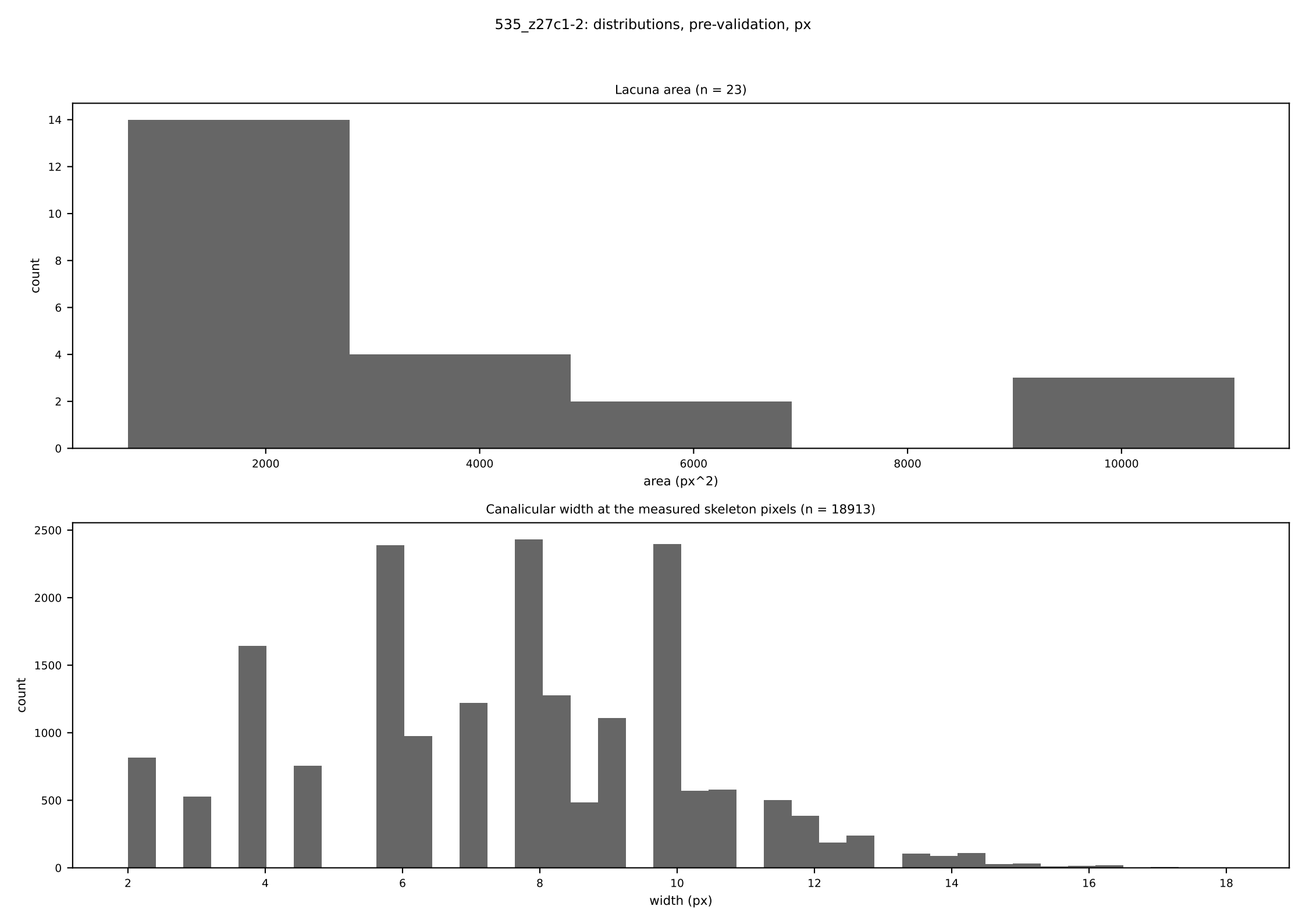
lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	254	405	319.664	604.529	8	10	8
2	168	244	243.652	614.357	4	7	8
3	122	122	153.368	267.38	5	4	2
4	162	228	230.095	376.978	4	7	5
5	55	65	124.61	391.718	2	5	7
6	224	543	320.108	931.725	6	8	11
7	90	99	161.853	406.576	3	4	6
8	85	107	98.9411	303.622	3	3	3
9	381	718	507.902	1193.57	14	15	15
10	298	631	369.907	837.683	9	10	9
11	157	331	186.823	522.002	5	5	6
12	24	24	100.296	163.752	1	2	1
13	139	231	199.61	432.433	4	5	8
14	174	232	281.007	525.245	6	7	5
15	97	162	169.61	327.25	3	5	3
16	107	184	157.297	299.279	4	5	4
17	482	888	560.659	1174.86	15	16	15
18	169	254	210.539	456.848	5	5	6
19	250	439	304.451	723.257	13	3	7
20	342	605	434.877	842.826	7	10	14
21	84	118	133.811	238.652	5	4	4
22	68	100	87.0122	169.953	2	2	3
23	375	800	454.848	1021.85	10	10	10


535_z27c1-2: distributions, pre-validation, px
Lacuna area (n = 23)
count
area (px^2)
0
2
4
6
8
10
12
14
2000
4000
6000
8000
10000
Canalicular width at the measured skeleton pixels (n = 18913)
count
width (px)
0
500
1000
1500
2000
2500
2
4
6
8
10
12
14
16
18